Independent Blinded Evaluation — Candidate Ranking

Generated: 2026-09-20 23:37:45 CDT
Noise level: 10.0% | Training interval: [0.0, 15.0] | Forecast interval: [15.5, 25.5]

Ranking follows the supplied instructions: (1) mathematical/biological robustness, (2) training-trajectory MSE, and (3) parsimony/interpretability. All candidates were treated as already accepted and were not refit, simulated, rejected, or otherwise modified.

Rank	Candidate ID	Main strength	Main concern	Reason for rank
1	C12	Smooth nonlinear saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.042481), then supplied complexity (11).
2	C21	Smooth quadratic saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.074198), then supplied complexity (16).
3	C18	Smooth quadratic saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.074198), then supplied complexity (14).
4	C27	Smooth polynomial saturation with no restricted domain.	Higher complexity without a corresponding training-MSE advantage.	Ranked using robustness first, then supplied training MSE (3.224947), then supplied complexity (19).
5	C24	Smooth polynomial saturation with no restricted domain.	Higher complexity without a corresponding training-MSE advantage.	Ranked using robustness first, then supplied training MSE (3.224947), then supplied complexity (17).
6	C15	Smooth polynomial saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.224947), then supplied complexity (12).
7	C22	Smooth polynomial saturation with no restricted domain.	Higher complexity without a corresponding training-MSE advantage.	Ranked using robustness first, then supplied training MSE (3.267176), then supplied complexity (17).
8	C13	Smooth polynomial saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.295519), then supplied complexity (12).
9	C16	Smooth quadratic saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.312241), then supplied complexity (14).
10	C19	Smooth quadratic saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.313552), then supplied complexity (16).
11	C10	Smooth nonlinear saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.324421), then supplied complexity (11).
12	C9	Smooth quadratic saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.494437), then supplied complexity (9).
13	C7	Smooth quadratic saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.540343), then supplied complexity (9).
14	C25	Smooth polynomial saturation with no restricted domain.	Higher complexity without a corresponding training-MSE advantage.	Ranked using robustness first, then supplied training MSE (3.577167), then supplied complexity (19).
15	C14	Smooth polynomial saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.743466), then supplied complexity (12).
16	C23	Smooth polynomial saturation with no restricted domain.	Higher complexity without a corresponding training-MSE advantage.	Ranked using robustness first, then supplied training MSE (3.753936), then supplied complexity (17).
17	C26	Smooth polynomial saturation with no restricted domain.	Higher complexity without a corresponding training-MSE advantage.	Ranked using robustness first, then supplied training MSE (3.802353), then supplied complexity (19).
18	C17	Smooth quadratic saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.850747), then supplied complexity (14).
19	C20	Smooth quadratic saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.852288), then supplied complexity (16).

Rank	Candidate ID	Main strength	Main concern	Reason for rank
20	C11	Smooth nonlinear saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (3.914992), then supplied complexity (11).
21	C8	Smooth quadratic saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (4.041614), then supplied complexity (9).
22	C3	Simple linear saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (4.055718), then supplied complexity (6).
23	C1	Simple linear saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (4.394288), then supplied complexity (6).
24	C2	Simple linear saturation with no restricted domain.	Training MSE is not the lowest among the robust polynomial candidates.	Ranked using robustness first, then supplied training MSE (4.599881), then supplied complexity (6).
25	C6	Stable positive equilibrium; good training fit among log candidates.	Logarithm imposes a finite domain and a nearby singular boundary.	Ranked using robustness first, then supplied training MSE (3.135465), then supplied complexity (8).
26	C4	Stable positive equilibrium; good training fit among log candidates.	Logarithm imposes a finite domain and a nearby singular boundary.	Ranked using robustness first, then supplied training MSE (3.530025), then supplied complexity (8).
27	C5	Stable positive equilibrium; good training fit among log candidates.	Logarithm imposes a finite domain and a nearby singular boundary.	Ranked using robustness first, then supplied training MSE (3.879829), then supplied complexity (8).

Top choice: C12

C12 is the highest-ranked candidate because it combines a non-restricted polynomial saturation form with the lowest supplied training-trajectory MSE in the accepted set. Its supplied complexity is moderate relative to the more complex candidates that do not provide a lower training MSE.

Comparison of the candidate set

The non-logarithmic candidates have no evident restricted-domain singularity in their expressions, so their ranking is primarily separated by the supplied training MSE and, for ties, complexity. The logarithmic candidates have an explicit finite-domain boundary, which creates a robustness concern under the first criterion and places them after the robust polynomial candidates despite some having competitive training MSE values.

Source trial information: Gaussian noise level 10.0%; training times 0, 3, 6, 9, 12, and 15 s; training volumes supplied in the accompanying TXT file.